

Machine-Checked Finite-Injury Priority Arguments: Friedberg–Muchnik and Sacks Splitting in Lean 4

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Abstract

We describe a Lean 4 formalization of two finite-injury priority arguments over a single, shared oracle-machine foundation: the Friedberg–Muchnik theorem, the 1950s solution to Post’s problem, which constructs two mutually nonreducible computably enumerable sets, and the Sacks Splitting Theorem in its splitting-with-non-reducibility form, which splits any non-computable c.e. set into two c.e. halves computing neither.

The development is *analytic* rather than synthetic: oracle computation is not a black-box semantic relation but an explicit language of oracle programs equipped with a step-indexed evaluator over finite oracle information. A halting bounded evaluation records both its output and a strict upper bound on the oracle positions it queried, and the central use theorem proves that such a computation is preserved under extra fuel and under oracle agreement below the recorded use. Restraint, injury, initialization, attention, eventual stability and effective stage construction are all represented directly, and computability of each stage function is discharged against Mathlib’s ordinary primitive- and partial-recursive infrastructure.

The second theorem is a test of the first’s foundation. The whole foundation transferred unchanged, and the recorded use—a design decision taken for Friedberg–Muchnik—turned out to be what bounds the restraint in Sacks, a theorem it was not designed for. The priority layer did not transfer at all: Friedberg–Muchnik’s injury argument counts discrete events, and a splitting requirement never acts, so a different induction is required. We report this as a negative result: “finite injury” names a family of arguments rather than a single reusable lemma. The two libraries are siblings, importing a common foundation and not each other, and the build enforces it.

Finally, we relate the local reducibility relation to Mathlib’s own `RecursiveIn` model by proving that every reduction there is realized by a program of our language. Since both conclusions are negations of reducibility, this is the direction that rules out a vacuously weak relation, and both theorems are restated over Mathlib’s notion. The development builds with no `sorry`, and every headline theorem depends only on `propext`, `Classical.choice` and `Quot.sound`.

1 Introduction

1.1 Two Kinds of Set

Call a set of natural numbers *decidable* if some algorithm, given a number, answers yes or no and always terminates. Call it *computably enumerable*, or c.e., if some algorithm lists its members — possibly forever, in no particular order, with no promise about when any given member appears. Every decidable set is c.e.; the converse fails, and the standard witness is the halting problem, the set of programs that eventually stop. One can list the halting programs by running all of them a step at a time and reporting each as it stops. One cannot decide membership, because a program that has not stopped yet may stop later, and nothing in the enumeration says which.

The asymmetry is the subject. A c.e. set carries *positive* information that accumulates: a number may join the set at some stage of the enumeration, and once it joins it never leaves. Negative information never becomes final. Every construction in this paper is a way of exploiting that asymmetry, and every difficulty in formalizing one comes from the same place.

To compare sets we relativize. Give an algorithm a free subroutine — an *oracle* — that answers membership questions about a set B . If some such algorithm decides A , write $A \leq_T B$: A is *Turing reducible* to B , meaning A is no harder than B . A halting oracle computation asks its oracle only finitely many questions, all below some bound called the *use* of that computation. This finiteness is the one lever the whole theory turns on, and we return to it below.

Reducibility sorts sets into layers of difficulty. At the bottom sit the decidable sets, reducible to everything. Near the top sits the halting problem, to which every c.e. set reduces.

1.2 Post’s Problem

In 1944 Post asked what lies between [7]. Is every c.e. set either decidable or as hard as the halting problem, or is there something strictly in the middle?

The question resisted for twelve years. It was settled in 1956–57, by Friedberg and by Muchnik independently [5, 6], and settled in a strong form: there are two c.e. sets A and B with $A \not\leq_T B$ and $B \not\leq_T A$. Neither can decide the other, so neither is decidable and neither is as hard as the halting problem. The middle is not merely occupied; it is occupied by incomparable things.

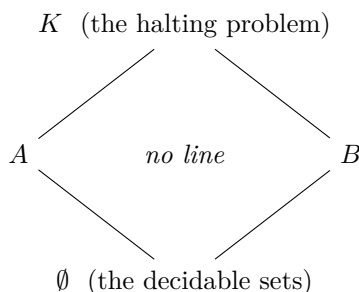


Figure 1: The classical picture behind Post’s problem: A and B sit strictly between the decidable sets and the halting problem, and neither reduces to the other. The formalization reported here proves the two missing edges — $A \not\leq_T B$ and $B \not\leq_T A$ — and proves that neither set is decidable. The halting problem itself does not appear in the development, so the vertical placement in this diagram is classical background rather than formalized content; see Section 1.4.

A second classical result concerns how a c.e. set can be taken apart. Sacks proved in 1963 that any c.e. set that is not decidable splits into two disjoint c.e. pieces, neither of which can decide the whole [8]. Nothing is thrown away — the pieces reassemble to the original — and yet each piece has strictly less power than the set it came from. Both halves of the difficulty are genuinely present, and neither half carries it alone.

1.3 Why These Proofs Are Hard to Formalize

Both results are proved by the same technique, introduced in the solution of Post’s problem and now called the *priority method*. It is worth saying what the technique does before saying why it resists formalization.

To make $A \not\leq_T B$, one must defeat every candidate reduction. Number the oracle programs $\Phi_0, \Phi_1, \dots$ and impose one requirement per program: R_e says that Φ_e with oracle B does not decide

A . Defeating a single Φ_e is easy in isolation. Pick a number x that nothing else cares about — a *fresh witness* — and keep it out of A . Watch Φ_e with oracle B run on input x . If it never converges, it never decides A and the requirement is met for free. If it converges and says “ $x \notin A$ ”, put x into A . Now the computation is wrong, and the requirement is met by contradiction.

The difficulty is that the computation was run against B *as it stood at that stage*, and B is still being enumerated. A later element entering B could change the computation and rescue it. Here the finite use is spent: the converging computation asked only finitely many questions of B , so it suffices to *restrain* B below that use — to promise that no number smaller than the use will ever be enumerated into B . The computation is then frozen for good, and the diagonalization holds.

But restraints conflict. A requirement that has promised to keep numbers out of B obstructs the requirements that want to put numbers in. The resolution is the priority ordering: requirements are ranked, and when a higher-ranked one acts, every lower-ranked one is *initialized* — its witness is discarded and its restraint cancelled, and it starts over with a fresh witness chosen above the new restraint. This is the *injury*. The proof then argues that injury is finite: a requirement is injured only by the finitely many stronger requirements above it, each of which acts only finitely often, so eventually every requirement is left alone forever and does its job.

This is a beautiful argument and an awkward one to make precise. Its content is carried by verbs in the future and past tense. *Choose a witness nothing else cares about* — nothing else meaning which requirements, and cares in what sense, at which stage? *Wait until the computation converges* — waiting is not an operation; the construction must at each stage make a decision from data it has, and “converges” is not such a datum, only “converges within the resources inspected so far”. *Restrain B below the use* — a promise about all future stages, made at one stage, which some later stage must be shown to keep. *Initialize everything weaker* — an edit to unboundedly many strategy records at once. *Eventually the stronger ones go quiet* — an induction along the priority order whose base case is that the strongest requirement is never injured at all, and whose step needs a measure of progress that no textbook states, because on paper it is obvious.

In a proof assistant none of this can stay in the tense system. Each phrase has to become a piece of data or a theorem about it. “Fresh” becomes a counter carried in the state and an invariant saying it dominates everything mentioned so far. “Waiting” becomes a decidable test on a bounded execution, plus a separate theorem that a computation which really does converge is eventually seen by that test. “Restraint” becomes a number in a record, plus a lemma that the enumeration respects it, plus the preservation theorem that converts respect into a frozen computation. “Initialize” becomes an explicit rewriting of the strategy list. “Eventually quiet” becomes an induction along priority carrying an explicit progress measure — and finding that measure is real work, since the informal proof supplies none.

That translation is the subject of this paper. We report a Lean 4 formalization of both theorems — Friedberg–Muchnik and Sacks splitting in its splitting-with-non-reducibility form — over one shared, explicitly finitary model of oracle computation, and we report what the translation cost and what it revealed.

1.4 What This Paper Claims

Our finding is not that the theorems are provable; they were proved seventy years ago. It is what happens when a second priority argument is built on the first one’s foundation. The foundation transferred completely, and one design decision made for the first theorem turned out to be the load-bearing one for the second. The priority layer transferred not at all, and we argue that no generalization of it would have. We report that as a negative result about reuse: “finite injury” names a family of arguments rather than a single reusable lemma. Section 2 is about both halves of

that finding.

Two scope notes belong here rather than at the end. First, what is formalized is incomparability together with non-computability: for the two constructed sets we prove $A \not\leq_T B$, $B \not\leq_T A$, and that neither is decidable. The classical step from there to “strictly between $\emptyset$ and the halting problem” uses the fact that every c.e. set reduces to the halting problem, and the halting problem, the jump, and completeness do not appear in the development at all. Figure 1 therefore shows more than we prove, and is labelled accordingly. Second, the full Sacks Splitting Theorem also makes the two halves *low*; that conclusion needs infinite-injury machinery and is deliberately out of scope. What we prove is the splitting-with-non-reducibility core.

2 The Formalization and Its Key Decisions

The development is four Lean libraries. `OracleComputability` fixes a machine model of oracle computation and proves the facts any priority argument over that model needs. `FriedbergMuchnik` and `SacksSplitting` are built on it, and neither imports the other; Lake enforces this, so building either from a clean tree compiles no module of the other. `PriorityArguments` is a single root module joining them. The two headline theorems are:

```
theorem friedberg_muchnik :
  exists A B : Set Nat,
    CE A /\ CE B /\ not (A <=T B) /\ not (B <=T A)

theorem sacks_splitting :
  forall A : Set Nat, CE A -> not (ComputableSet A) ->
    exists A0 A1, CE A0 /\ CE A1 /\ Disjoint A0 A1
      /\ A0 union A1 = A
      /\ not (A <=T A0) /\ not (A <=T A1)
```

The vocabulary — `CE`, `ComputableSet`, and the relation displayed as $\leq_T$ — is defined in the shared library and imported by both clients, not redefined, so the two theorems are stated in one language.

2.1 Deciding What Is Primary

A finitary oracle model. The development does not start from a semantic type of oracle functionals. Doing so would require an oracle-machine model, continuity or compactness theorems, extraction of finite uses from semantic computations, and a connection between functionals and executable indices — all before the first requirement could be stated. Instead *finite execution is primary*. The evaluator runs a concrete program for bounded fuel against a partial oracle $\mathbb{N} \rightarrow \text{Option Bool}$, and returns one of three outcomes: it halted, recording both its output and the use; it got *stuck*, having asked a question the finite oracle could not answer; or it ran out of fuel. Distinguishing the last two matters, because they are repaired by different things — more oracle information, or more resources — and a priority construction reacts to them differently. Infinite computation is then defined as existence of a finite successful run against a total oracle.

Recording the use inside the result, rather than recovering it afterwards by a continuity argument, is the decision the rest of the development rests on. It makes the use principle a statement about data the construction can inspect at a stage. The principle itself, `run_halt_mono`, says that a halting bounded run survives more fuel and any oracle that preserves the answers it actually used. The hypothesis is one-sided and deliberately weak: positions at or above the recorded use are

unconstrained, and so are positions below the use where the original oracle had no information. That weakness is exactly what a restraint can deliver.

Finite sets, not Boolean prefixes. Evolving c.e. approximations are monotone finite sets, with Boolean lists appearing only as bounded snapshots supplied to one computation:

```
def StageMono (F : Nat -> Finset Nat) : Prop :=
  forall s, F s subset F (s + 1)

def limitSet (F : Nat -> Finset Nat) : Set Nat :=
  {n | exists s, n in F s}
```

The alternative — representing an approximation as a prefix-monotone Boolean string, where each bit is fixed once it appears — is not merely inconvenient, it proves the wrong theorem. Fixing bits makes the limit sets decidable, and the development proves that a decidable set reduces to every oracle. A prefix-monotone construction would therefore have handed back precisely the reduction the theorem denies. The two representations coexist because they carry different information: a stage set may accumulate new positive information at any later stage, while a snapshot merely answers finitely many questions for one bounded run.

Where Mathlib enters, and where it does not. Ordinary, non-oracle computability is imported; oracle computability is built locally. Exactly three files import Mathlib’s computability library: one for the primitive-recursive closure theorems used to show the stage functions computable, one embedding Mathlib’s partial-recursive codes into the query-free fragment of our language, and one relating our reducibility to Mathlib’s own oracle model. The local oracle semantics — the program syntax, the evaluator, the use principle, the reducibility vocabulary, the approximation layer — mention Mathlib’s codes nowhere, so changes to Mathlib’s internal coding decisions cannot reach them; the clients meet Mathlib only where they must, in discharging computability of their stage functions. This boundary lets the construction reason about explicit finite uses while still reusing Mathlib’s closure library [2, 1].

Reducibility vocabulary. The central definition is that $A \leq_T B$ holds when some local oracle program computes the characteristic function of A , on every input, from the total characteristic function of B :

```
def TuringReducible (A B : Set Nat) : Prop :=
  exists c : OracleCode,
    forall x : Nat,
      Computes c (charFun B) x (natOfBool (charFun A x))
```

Reflexivity, transitivity (by substituting one program for every query node of another), and the fact that decidable sets reduce to every oracle are all proved, so the relation is a preorder with the expected bottom behaviour. The program numbering is proved to be a bijection between $\mathbb{N}$ and the code type, which is what turns “for every index e ” into “for every reduction”: the final diagonalization takes an arbitrary candidate program, passes to its index, and lets the corresponding requirement kill it.

2.2 What Transferred to the Second Theorem, and What Did Not

The foundation was built for one theorem. To test whether it was a foundation rather than scaffolding, we formalized a second finite-injury theorem against it and recorded, component by

component and before writing any of the second construction, what we expected to reuse.

The foundation transferred whole. Every component of the machine model, the use principle, the reducibility vocabulary, the Mathlib bridge and the approximation layer was imported and used unchanged. Two items are worth naming.

The restraint lemma — a computation seen halting against a stage snapshot survives to the limit provided nothing below its recorded use is enumerated later — was already stated generically in the stage sequence rather than in terms of the first construction’s sets. The preservation argument at the heart of Sacks splitting therefore reuses it verbatim. This was the open question the reuse inventory was written to answer, and the answer is clean.

More interestingly, the determinism lemma `run_halt_unique` carries more weight in the second theorem than in the first. In Friedberg–Muchnik it says only that a computation has one output. In Sacks it is what makes *the use of the limit computation at y* a well-defined number, and that number is what bounds the restraint function. Recording the use rather than only the output was a decision taken for the first theorem, for reasons internal to it; it paid for a theorem it was not designed for. That is the strongest evidence we have that the foundation generalizes.

The priority layer transferred not at all. The Friedberg–Muchnik injury argument counts *events*. A requirement’s record carries a rank in $\{0, 1, 2\}$ — has it a witness, has it acted — receiving attention raises the rank strictly, and only injury from above resets it. So once the stronger requirements go quiet, a requirement acts at most twice more. This is the explicit progress measure the informal proof does not supply, and it is small enough that the induction along priority is arithmetic.

A splitting requirement generates no events. It never acts: the elements are not ours to choose, they arrive from the enumeration of the given set, and the construction only decides which half each one joins. Its restraint moves continuously with the length of agreement between the reduction and the set being split, so there is no discrete step to count and no rank to raise. The replacement is a three-part simultaneous induction along the priority order: bounded restraints below j give finite injury to j ; finite injury to j gives satisfaction of j ; satisfaction of j bounds j ’s restraint.

The two arguments share a shape — induction along the priority order, no closed-form injury bound — and no content. The two files have no declaration in common. We looked for a common generalization worth stating and did not find one, and we record that as a negative result about reuse rather than as a defect of the first development’s design: “finite injury” names a family of arguments, not a single lemma one can prove once and instantiate.

One further difference is forced by the mathematics rather than by taste. The splitting restraint must be *cumulative*, kept in the state as a running maximum. The raw stage restraint is not monotone — the length of agreement drops whenever a number below it enters the set — and a restraint that can drop protects nothing, since a computation frozen at one stage could be destroyed later merely because its requirement had temporarily stopped asking. Friedberg–Muchnik needs no such device, its restraint being written once, when a requirement acts.

Where the hypothesis is spent. Sacks splitting takes a hypothesis Friedberg–Muchnik does not have: the set being split is not decidable. The step that consumes it is the one place where the second development required an argument with no analogue in the first. If a requirement is never injured after some stage and its reduction nevertheless computes the characteristic function of A everywhere, then A is decidable: to decide $y \in A$, search for a stage past the last injury whose length of agreement exceeds y , and report whether y is in the approximation at that stage. Such a stage

exists because the reduction is total; the answer is correct because past the last injury the agreeing computation is frozen by the restraint lemma, so it *is* the true value; and the search is effective because the whole construction is primitive recursive. Since A is not decidable, no requirement can be in that situation, so every requirement is satisfied.

Friedberg–Muchnik has no counterpart. Its requirements are satisfiable because their witnesses are fresh, which is a property of the construction; a splitting requirement is satisfiable only because of a property of the *given* set. We describe this as a step of the classical proof made explicit rather than as new mathematics: it is the point at which a textbook appeal to non-computability becomes a decision procedure that has to be exhibited and shown effective.

What the second theorem cost. The shared foundation is roughly 2,870 lines; the Friedberg–Muchnik client is roughly 2,000 and the Sacks client roughly 1,550. The ratio is the point: the expensive part — model, evaluator, use principle, Mathlib bridge, approximation layer — was already present, and the second theorem paid only for its own strategy.

Two components of the first development turned out to be artifacts of its own design rather than necessities. Its ten-field state invariant exists to keep freely chosen witnesses fresh, distinct, and clear of higher-priority restraints; a construction that does not choose its elements needs none of it, and the two invariants that remain — the halves stay disjoint, and their union is the current stage of the given set — are one induction. And its computability layer needs a shadow implementation on plain tuples, because its state types are custom structures with no `Primcodable` instance; defining the splitting state as a tuple from the outset removes that layer, its simulation theorem, and its list-surgery lemmas entirely.

The exercise also found two defects that only a second client could find. Three generic helpers about primitive recursion had been declared `private` inside the first development’s computability file. Nothing in them mentions either construction — they are what any stage function needs in order to be shown computable — but `private` made them invisible downstream. Nothing needed generalizing; they needed exporting. Separately, the first development’s root module imported a file that did not exist, so a clean checkout did not build at all. Its own build never noticed, because the file was present locally. Both are invisible until someone tries to build on the code.

2.3 Analytic, and Bridged Rather Than Asserted

Post’s problem has been mechanized before, and the contrast is a design decision rather than a survey obligation. Zeng, Forster and Kirst formalize a solution in Coq [9], building on the synthetic oracle computability of Forster, Kirst and Mück [4]. Their setting is *synthetic*: one works in the Calculus of Inductive Constructions and takes every function $\mathbb{N} \rightarrow \mathbb{N}$ to be computable by axiom, so there is no object-level machine model and none of the details it brings. That is a deliberate and effective choice, and it answers the standard complaint that explicit models of computation drown a computability proof in bookkeeping that stays invisible on paper. Two points of their work bear directly on ours: the solution they formalize is Soare’s low simple set, which their abstract explicitly contrasts with “the full Friedberg–Muchnik theorem constructing two incomparable predicates”; and the same abstract records that Bauer posed a synthetic proof of Friedberg–Muchnik as a challenge.

The present development is *analytic*, and pays the cost the synthetic approach avoids. What is bought is that every informal appeal to “preserve the computation below its use” becomes an application of one lemma about one evaluator, and that the resulting reducibility relation can be compared with an external one by proof rather than by fiat. Neither approach dominates; they answer different questions about the same theorem.

That comparison is the last design decision. Mathlib contains a model of oracle computability [3]: an inductive relation `RecursiveIn` on partial functions, Turing reducibility and equivalence, and degrees as the quotient with a partial-order instance. It does not contain a use principle, oracle-relative step-indexed evaluation, oracle-relative computable enumerability, or the jump, and it is not connected to Mathlib’s computably enumerable predicates, which live in a separate file developed for the unrelativized theory. The two libraries are complementary: Mathlib has the degree *structure* we do not build, and we supply the finite-computation machinery it lacks.

So we prove the connection instead of assuming it. A bridge file establishes that every reduction in Mathlib’s model is realized by a program of our language, by induction over `RecursiveIn` with one program per constructor. This is the direction that matters: both of our conclusions are *negations* of reducibility, so the risk to guard against is that our relation admits too few reductions and thereby makes the negations cheap. It does not, and both theorems are consequently restated over Mathlib’s relation. The converse inclusion is not needed for these results and is not proved.

Two priority claims, stated with the caution they deserve. We are not aware of another formalization of Post’s problem in an analytic setting with an explicit oracle machine model, nor of another in Lean; the synthetic precedent above is the closest. We are also not aware of any formalization of the Sacks Splitting Theorem in any proof assistant. Both claims rest on a literature and repository search rather than an exhaustive one, and neither is load-bearing for the technical content of this paper.

2.4 The Shape of the Implementation

The details of the machinery are tabulated in Appendix A; what follows is the shape.

The program language is Kleene’s schemata — zero, successor, the two projections, pairing, composition, primitive recursion and unbounded search — plus one oracle primitive, `query`, which asks whether its input is in the oracle set. A code with no `query` node is an ordinary partial recursive program; the bridge embeds every Mathlib partial-recursive code into that fragment and proves the embedding semantics-preserving, which is the direction the construction consumes. The evaluator’s fuel discipline deliberately copies Mathlib’s `evaln`, which is what lets the embedding be proved by a single structural induction rather than by a simulation with fuel translation. The whole evaluator is proved primitive recursive, in the form in which the construction consumes it: numbered codes in, encoded results out.

The Friedberg–Muchnik construction keeps two enumeration lists, one record per requirement holding a witness, an acted flag and a restraint, and a freshness counter dominating everything mentioned so far. At each stage the least requirement requiring attention receives it: with no witness it is given the counter’s value; with an unacted witness whose diagonalizing computation has appeared within the stage’s resources, it acts — enumerating the witness, recording the computation’s use as its restraint, and resetting every weaker record. A hypothesis-free case analysis proves that every stage is exactly one of idle, appointment, or action, so no execution path can escape the verification. Ten invariants, proved together by induction over those three transitions, carry the properties the satisfaction proof cites by name, of which the load-bearing one says that an acted requirement’s diagonalizing computation still halts against the current stage.

The splitting construction is smaller for a structural reason: it makes no positive moves. A stage takes the numbers the given set’s own enumeration has just produced and routes each to the half not being used as an oracle by the strongest requirement that restrains it, defaulting to the first half when nothing restrains it. Two invariants suffice: the halves stay disjoint, and their union is the current stage of the given set.

The computable-enumerability obligation is discharged the same way in both developments and

is the most expensive part of each. The stage function is shown primitive recursive using Mathlib’s closure library, an unbounded search over stages turns it into a partial recursive semi-decider, and the bridge converts that into the project’s CE. For Friedberg–Muchnik this requires the shadow tuple implementation and a simulation theorem tying it to the real machine, and it dominates the build: one file accounts for roughly seven eighths of the project’s compile time. The instructive failure along the way was assembling that derivation as a single large combinator term, which made failed unifications fall back into the pairing encodings and left the elaborator symbolically evaluating arithmetic for tens of millions of heartbeats. Confining each tuple-sized comparison to its own lemma fixed it.

3 Conclusion

Two classical finite-injury priority arguments are formalized in Lean 4 over one explicitly finitary model of oracle computation. The informal temporal reasoning of the textbook proofs — choose a fresh witness, wait for convergence, restrain below the use, initialize what is weaker, argue that the stronger requirements eventually go quiet — becomes finite executable stages, explicit state components, recorded uses, preserved invariants, a bounded progress measure, and an induction along priority. The constructions are not merely type-checked but executable: the stage machines run, and produce the witnesses, restraints and enumerations the proofs reason about.

The methodological lesson is about representation. Priority arguments are not beyond current provers, but they require representations that expose finite information at exactly the points where the informal proof appeals to it. Recording the use inside the evaluator’s result, rather than recovering it after the fact, is what makes restraint a manipulable object; keeping approximations as monotone finite sets rather than prefix-fixed Boolean strings is what keeps the limit sets from being decidable and the theorem from being false.

The second lesson is the one we did not expect, and it is negative. A foundation built for one priority argument served a second one completely — including a lemma whose importance in the second theorem exceeds its importance in the first — while the priority layer itself transferred not at all, and no generalization of it appears to be worth stating. “Finite injury” names a family of arguments related by shape, not a lemma one proves once. A useful reusable priority framework, if there is one, will have to abstract requirements, strategies, priority order, initialization and progress measures at a level neither of these two developments currently occupies; we regard that as open rather than as routine engineering.

The natural next target is infinite injury. It is what the omitted conclusion of the Sacks Splitting Theorem needs — lowness of the two halves — and it is the first place where the finite progress measure that made the first theorem tractable is unavailable in principle rather than by accident of design. The other loose ends are smaller. Transitivity of the local reducibility relation, listed as future work in an earlier draft of this development, is now proved by query substitution, so the relation is a preorder; what remains is that the bridge to Mathlib’s oracle model is proved in one direction only, which suffices for the negative results here but not for a general transfer, and that no degree structure — the quotient, the jump, completeness — is formalized, so the classical reading of these theorems as statements about degrees remains external to the artifact.

References

- [1] Mathlib documentation: Mathlib.computability.partrec. https://leanprover-community.github.io/mathlib4_docs/Mathlib/Computability/Partrec.html, 2026. Accessed 2026-07-

- [2] Mario Carneiro. Formalizing computability theory via partial recursive functions. In *10th International Conference on Interactive Theorem Proving (ITP 2019)*, volume 141 of *Leibniz International Proceedings in Informatics*, pages 12:1–12:17. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2019.
- [3] Tanner Duve and Elan Roth. Mathlib.Computability.TuringDegree, 2025. Mathlib4 library file; `RecursiveIn`, `TuringReducible`, `TuringDegree`.
- [4] Yannick Forster, Dominik Kirst, and Niklas Mück. Oracle computability and Turing reducibility in the calculus of inductive constructions. *CoRR*, abs/2307.15543, 2023.
- [5] Richard M. Friedberg. Two recursively enumerable sets of incomparable degrees of unsolvability (solution of post’s problem, 1944). *Proceedings of the National Academy of Sciences of the United States of America*, 43(2):236–238, 1957.
- [6] A. A. Muchnik. On the unsolvability of the problem of reducibility in the theory of algorithms. *Doklady Akademii Nauk SSSR*, 108:194–197, 1956. In Russian.
- [7] Emil L. Post. Recursively enumerable sets of positive integers and their decision problems. *Bulletin of the American Mathematical Society*, 50:284–316, 1944.
- [8] Gerald E. Sacks. On the degrees less than $0'$. *Annals of Mathematics*, 77:211–231, 1963.
- [9] Haoyi Zeng, Yannick Forster, and Dominik Kirst. Post’s problem and the priority method in CIC. In *30th International Conference on Types for Proofs and Programs (TYPES)*, 2024. Synthetic formalization in Coq.

A Textbook-to-Lean Map

The tables below name, for each concept of the informal proofs, the Lean declaration that carries it and the file it lives in. File paths are relative to the repository root. Together with Appendix B they replace the file-by-file walkthrough that would otherwise occupy the body of the paper.

A.1 Shared foundation: `OracleComputability/`

Concept	Declaration	File
Oracle program	OracleCode	OracleCode.lean
Partial oracle	PartOracle, ofFun, ofSnapshot	FiniteEval.lean
Bounded computation	run	FiniteEval.lean
Output and use	HaltData	FiniteEval.lean
Convergence / no information / no fuel	RunResult	FiniteEval.lean
Input guard	run_halt_input_lt	FiniteEval.lean
Use principle	run_halt_mono	Use.lean
More fuel is harmless	run_halt_fuel_mono	Use.lean
More oracle information is harmless	run_halt_oracle_ext	Use.lean
Determinism across oracles	run_halt_unique	Use.lean
Infinite computation	Computes	InfiniteEval.lean
One output per input	Computes.unique	InfiniteEval.lean
Finite $\leftrightarrow$ infinite	computes_iff_initialSegment	InfiniteEval.lean
Program numbering	encodeCode, ofNatCode	Numbering.lean
Enumeration adequacy	ofNatCode_encodeCode	Numbering.lean
Numbering is onto $\mathbb{N}$	encodeCode_ofNatCode	Numbering.lean
Turing reducibility	TuringReducible	Reducibility.lean
Computable enumerability	CE	Reducibility.lean
Decidable set	ComputableSet	Reducibility.lean
Characteristic function	charFun	Reducibility.lean
$\leq_T$ is reflexive	TuringReducible.refl	Reducibility.lean
Decidable reduces to everything	ComputableSet.turingReducible	Reducibility.lean
Query substitution	subst	Composition.lean
$\leq_T$ is transitive	TuringReducible.trans	Composition.lean
Ordinary code embedding	embed, run_embed	MathlibBridge.lean
Mathlib functions realized	partrec_realized	MathlibBridge.lean
Semi-decider gives CE	ce_of_partrec_dom	MathlibBridge.lean
Mathlib's oracle relation	MathlibTuringReducible	MathlibOracleBridge.lean
Realization of a Mathlib reduction	LocallyRealizes	MathlibOracleBridge.lean
Induction over RecursiveIn	exists_code_of_recursiveIn	MathlibOracleBridge.lean
Transfer theorem	turingReducible_of_mathlib	MathlibOracleBridge.lean
Negative results transfer	not_mathlibTuringReducible	MathlibOracleBridge.lean
Evaluator on numbered codes	nrun	RunPrimrec.lean
Evaluator is primitive recursive	nrun_primrec	RunPrimrec.lean
Least index with a property	range_find?_least	Priority.lean
c.e. approximation	StageMono	Approximation.lean
Limit set	limitSet	Approximation.lean
Settling below a bound	StageMono.stabilizesBelow	Approximation.lean
Oracle snapshot	snapshot, snapshotOf	Approximation.lean
Restraint preserves a computation	run_halt_limit_of_restraint	Approximation.lean
A true computation becomes visible	run_halt_snapshot_of_limit	Approximation.lean
Generic Primrec helpers	primrec_find?, primrec_memDecide, primrec_snapshotOf	PrimrecTools.lean

Table 1: The shared foundation.

A.2 Friedberg–Muchnik: FriedbergMuchnik/

Concept	Declaration	File
Strategy record	ReqState	Construction.lean
Witness, acted flag, restraint	witness, acted, restraint	Construction.lean
Construction state	ConsState	Construction.lean
Bounded convergence test	convCheck	Construction.lean
Requires attention	requiresAttention	Construction.lean
Appointment and action	stepAt	Construction.lean
One stage	stepState	Construction.lean
The construction	stageState	Construction.lean
Stage approximations	AstageF, BstageF	Construction.lean
The two sets	Aset, Bset	Construction.lean
Enumeration is monotone	AstageF_mono, BstageF_mono	Construction.lean
The attended requirement	attended	StageDynamics.lean
Priority discipline	attended_least	StageDynamics.lean
Idle / appoint / act trichotomy	step_cases	StageDynamics.lean
No injury upward	reqs_getElem?_succ_of_lt	StageDynamics.lean
Only an action enumerates	Alist_succ, Blist_succ	StageDynamics.lean
Freshness never decreases	fresh_mono	StageDynamics.lean
The ten invariants	ConsInv	Invariants.lean
Invariants hold at every stage	consInv_stage	Invariants.lean
Preserved computation	ConsInv.acted_comp	Invariants.lean
Witnesses clear higher restraints	ConsInv.wit_ge	Invariants.lean
Progress measure	rank, reqRank	FiniteInjury.lean
Rank is bounded by 2	rank_le_two	FiniteInjury.lean
Attention raises rank	reqRank_incr	FiniteInjury.lean
Quiet interval freezes a record	record_stable_of	FiniteInjury.lean
Finite injury	quiet_above	FiniteInjury.lean
Stabilization package	exists_stable	FiniteInjury.lean
Stable means never asks again	stable_not_requires	FiniteInjury.lean
Restraint is respected	restraint_respected_A/B	Requirements.lean
Requirements satisfied	R_satisfied, S_satisfied	Requirements.lean
The two non-reducibilities	not_A_le_B, not_B_le_A	Requirements.lean
Shadow implementation	St, RS, encSt, stageN	CE.lean
Simulation theorem	encSt_stage	CE.lean
The sets are c.e.	ce_Aset, ce_Bset	CE.lean
The theorem	friedberg_muchnik	Main.lean
Neither set is decidable	Aset_not_computable, Bset_not_computable	Main.lean

Table 2: The first priority argument.

A.3 Sacks splitting: SacksSplitting/

Concept	Declaration	File
The given set as watchable stages	<code>enumStage, enumSet</code>	<code>Basic.lean</code>
CE gives such an enumeration	<code>ce_eq_enumSet</code>	<code>Basic.lean</code>
Newly arrived elements	<code>newAt</code>	<code>Basic.lean</code>
Primitive-recursive filtering	<code>filterB</code>	<code>Basic.lean</code>
Construction state (a tuple)	<code>State</code>	<code>Requirements.lean</code>
Which half a requirement watches	<code>oracleOf</code>	<code>Requirements.lean</code>
Bounded agreement test	<code>agreeAt</code>	<code>Requirements.lean</code>
Length of agreement	<code>lenAgree</code>	<code>Requirements.lean</code>
Stage restraint	<code>restraintAt</code>	<code>Requirements.lean</code>
Cumulative restraint	<code>newRestrains, Rest</code>	<code>Construction.lean</code>
Routing rule	<code>routeTo</code>	<code>Construction.lean</code>
One stage	<code>step, stageState</code>	<code>Construction.lean</code>
The two halves	<code>halfList, halfF, halfSet</code>	<code>Construction.lean</code>
Halves are disjoint	<code>disjoint_halfSet</code>	<code>Construction.lean</code>
Halves reassemble the set	<code>union_halfSet</code>	<code>Construction.lean</code>
Construction is computable	<code>primrec_stageState</code>	<code>CE.lean</code>
The halves are c.e.	<code>ce_halfSet</code>	<code>CE.lean</code>
The decision procedure	<code>computableSet_of_agreement</code>	<code>CE.lean</code>
Injury comes from above	<code>injured_by_higher</code>	<code>FiniteInjury.lean</code>
Bounded below $\Rightarrow$ finite injury	<code>noInjury_of_bounded</code>	<code>FiniteInjury.lean</code>
Restraint is respected	<code>restraint_respected</code>	<code>FiniteInjury.lean</code>
Where the hypothesis is spent	<code>unsatisfied_requirement_computes</code>	<code>FiniteInjury.lean</code>
Use of the limit computation	<code>limUse</code>	<code>FiniteInjury.lean</code>
Satisfied $\Rightarrow$ bounded restraint	<code>restraint_bounded_of</code>	<code>FiniteInjury.lean</code>
The priority induction	<code>restraint_bounded</code>	<code>FiniteInjury.lean</code>
Requirements satisfied	<code>requirement_satisfied</code>	<code>FiniteInjury.lean</code>
Neither half computes the set	<code>not_reducible_half</code>	<code>Main.lean</code>
The theorem	<code>sacks_splitting</code>	<code>Main.lean</code>
The split is non-degenerate	<code>halfSet_ne</code>	<code>Main.lean</code>

Table 3: The second priority argument.

A.4 Where the two meet: PriorityArguments.lean

Concept	Declaration
Sacks splitting is not vacuous	<code>friedbergMuchnik_set_splits</code>
Friedberg–Muchnik over Mathlib’s model	<code>friedberg_muchnik_mathlib</code>
Sacks splitting over Mathlib’s model	<code>sacks_splitting_mathlib</code>

Table 4: The only declarations depending on both priority arguments.

B Verified Artifact Facts

Every figure below was measured against the repository, on a clean build, rather than asserted. Measurements are from an Apple M5 (10 cores, 32 GB) with Mathlib’s compiled artifacts supplied

by `lake exe cache get`.

Item	Value
Lean toolchain	leanprover/lean4:v4.26.0
Mathlib version	v4.26.0
Mathlib revision	2df2f0150c275ad53cb3c90f7c98ec15a56a1a67
Lake libraries	4
Lean files	30
Lines of Lean	6,499
Lines excluding comments and blanks	4,471
Hand-written <code>theorem</code> declarations	274
Hand-written <code>def/abbrev</code> declarations	90
<code>lake build</code>	succeeds, 829 jobs
Clean build, Mathlib cached	2 min 2 s
Slowest module (<code>FriedbergMuchnik/CE.lean</code>)	106 s
Declarations checked with <code>#print axioms</code>	37
Axiom footprint, all 37	<code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code>
Axioms declared by the development	0
<code>sorry / admit</code>	0
<code>native_decide / unsafe / partial def</code>	0
<code>noncomputable</code> declarations	2
Declarations with raised <code>maxHeartbeats</code>	11
Repository license	none
CI workflow	none

Table 5: Artifact facts, measured at the revision described in Appendix C.

Library	Files	Lines	Code lines
<code>OracleComputability</code>	14	2,866	2,000
<code>FriedbergMuchnik</code>	8	2,002	1,494
<code>SacksSplitting</code>	7	1,548	948
<code>PriorityArguments</code>	1	83	29

Table 6: Size by library. “Code lines” excludes comments and blanks.

The two `noncomputable` declarations are the classical characteristic function `charFun`, which appears only in the statements of the reducibility vocabulary, and `limUse`, which appears only in the verification of the second theorem. Neither occurs in a construction. The raised heartbeat budgets are elaboration limits in the two computable-enumerability files and cannot affect the axiom footprint, which is the ground truth that no logical shortcut was taken.

The constructions run. Both stage machines are executable Lean functions, and evaluating them is a stronger check than the absence of `sorry`: it shows the definitions are not vacuously satisfiable stubs. Evaluating the Friedberg–Muchnik machine — the one the priority proof reasons about, not the shadow implementation used for the computability proof — at stage 20 gives enumerations $\{12, 8, 6, 4, 0\}$ and $\{13, 10, 1\}$ for the two sets, a freshness counter of 14, and strategy records in which requirements 8 and 10 hold restraints 9 and 11 respectively. Evaluating the splitting machine on the code that halts everywhere routes the first eight elements into a single half, all restraints remaining zero, since a reduction that always answers zero never agrees with a set that is accumulating members.

C Artifact Reproduction

The repository is available at:

`https://github.com/aaslyan/friedberg-muchnik-lean`

A fresh build:

```
git clone https://github.com/aaslyan/friedberg-muchnik-lean
cd friedberg-muchnik-lean
lake exe cache get
lake build
```

The first command fetches Mathlib’s compiled artifacts; on a cold machine it downloads several gigabytes. With them present, `lake build` compiles the four project libraries in about two minutes and reports `Build completed successfully (829 jobs)`.

To check the two theorems and their axiom footprints:

```
cat >/tmp/check.lean <<'EOF'
import PriorityArguments

#check FriedbergMuchnik.friedberg_muchnik
#check SacksSplitting.sacks_splitting
#print axioms FriedbergMuchnik.friedberg_muchnik
#print axioms SacksSplitting.sacks_splitting
#print axioms PriorityArguments.friedberg_muchnik_mathlib
#print axioms PriorityArguments.sacks_splitting_mathlib
EOF

lake env lean /tmp/check.lean
```

This reports the two theorem statements followed by four identical axiom lines:

```
'FriedbergMuchnik.friedberg_muchnik' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'SacksSplitting.sacks_splitting' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'PriorityArguments.friedberg_muchnik_mathlib' depends on axioms:
  [propext, Classical.choice, Quot.sound]
'PriorityArguments.sacks_splitting_mathlib' depends on axioms:
  [propext, Classical.choice, Quot.sound]
```

To confirm that the libraries are siblings — that neither priority argument depends on the other — build either alone from a clean tree and observe that no module of the other is compiled:

```
lake build SacksSplitting
```

To search for proof escape hatches across the project sources:


```
grep -rnE "sorry|admit|native_decide|unsafe|^axiom |partial def" \
  --include='*.lean' \
  OracleComputability FriedbergMuchnik SacksSplitting \
  PriorityArguments.lean
```

This produces no output. To run the constructions rather than only type-check them:

```
cat >/tmp/exec.lean <<'EOF'
import PriorityArguments

#eval (FriedbergMuchnik.stageState 20).Alist
#eval (FriedbergMuchnik.stageState 20).Blist
#eval SacksSplitting.enumStage 0 8
#eval (SacksSplitting.stageState 0 8).1
EOF

lake env lean /tmp/exec.lean
```

The repository additionally carries a fact dossier, `paper/DOSSIER.md`, recording the evidence behind each claim in this paper — what was read from source, what was measured, what was verified against primary literature, and what remains open.